Project Title: Customer Shopping Trends and Behavior Analysis

Problem Statement

Retail businesses collect information about purchases, product categories, prices, discounts, payment methods, and customer preferences. Without analyzing this data, it can be difficult to understand buying patterns and improve sales strategies. This project will analyze customer shopping behavior and present key trends through visualizations and an interactive dashboard.

Target Users

- **Retail managers:** Monitor sales trends and identify popular products.
- **Marketing teams:** Understand customer preferences and plan promotions.
- **Business owners:** Make data-informed decisions about products, pricing, and inventory.
- **Analysts:** Explore customer segments, purchase frequency, and spending patterns.

What We Will Do

Step	Plan
Collect data	Use a public or approved dataset containing purchases, product details, and customer attributes.
Clean data	Handle missing values, duplicates, and inconsistent categories.
Analyze trends	Calculate sales, average spend, purchase frequency, and popular categories.
Visualize findings	Compare spending by category, season, age group, and payment method where available.
Build dashboard	Create interactive filters for product category, season, customer segment, and purchase type.
Recommend actions	Summarize insights that may help improve promotions, inventory planning, and customer experience.

Expected Dashboard

- KPI cards: total sales/revenue, number of purchases, average purchase value, and number of customers (if available).
- Bar chart: sales or purchase count by product category.
- Line chart: sales trends over time or across seasons.

- Donut chart: share of purchases by payment method or product category.
- Stacked bar chart: compare shopping patterns across customer segments.
- Filters: date/season, product category, customer segment, and payment method.

Tools and Technologies

Python with Pandas and NumPy for cleaning and analysis; **Matplotlib/Seaborn** for exploratory charts; **Power BI or Tableau** for an interactive dashboard; **Excel** for basic data inspection.

Expected Outcomes

- Identify best-selling categories and common purchase patterns.
- Understand spending and purchase frequency across available customer segments.
- Explore seasonal trends and payment preferences.
- Provide data-based recommendations for promotions, stock planning, and customer experience.

Expected dashboard



Summary

This project analyzes customer shopping data to identify purchasing trends, product preferences, spending patterns, and seasonal behavior. The findings will be presented in an interactive dashboard to support better retail and marketing decisions. Conclusions will be limited to patterns supported by the dataset.